

DevOps Lab Task - Program 3: Dockerized Python Flask Application

This document demonstrates how to containerize a Python Flask application using Docker.

The process includes creating a Flask app, defining dependencies, writing a Dockerfile, building an image, and running the container to serve the Flask web app inside Docker.

Dockerfile Used in this Program

```
FROM python:3.10-slim

WORKDIR /app

COPY req.txt /app

RUN pip install -r req.txt

COPY . .

EXPOSE 8080

CMD ["python", "app.py"]
```

Steps to Build and Run the Flask Application in Docker

Step 1: List the project files

```
$ ls
app.py Dockerfile req.txt
```

Step 2: Build the Docker image

```
$ sudo docker build -t flask-image .
```

Step 3: Run the Flask container and map ports

```
$ sudo docker run -p 8080:8080 --name flask-cont flask-image
```

Step 4: Verify the Flask app in the browser

Navigate to <http://localhost:8080> - it should display "Hello, Flask is running inside Docker!"

Step 5: List the running containers

\$ sudo docker ps

```
Session  Actions  Edit  View  Help

└─$ ls
app.py  Dockerfile  req.txt

└─(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab3]
└─$ sudo docker build -t falsk:2.0 .
[+] Building 2.5s (10/10) FINISHED
⇒ [internal] load build definition from Dockerfile
⇒ ⇒ transferring dockerfile: 169B
⇒ [internal] load metadata for docker.io/library/python:3.10-slim
⇒ [internal] load .dockerignore
⇒ ⇒ transferring context: 2B
⇒ [1/5] FROM docker.io/library/python:3.10-slim@sha256:2a8c7d9efdae9b636de80a187a4ce31374ffb3883a7be06232d25a96e8538bf4
⇒ [internal] load build context
⇒ ⇒ transferring context: 288B
⇒ CACHED [2/5] WORKDIR /app
⇒ CACHED [3/5] COPY req.txt /app
⇒ CACHED [4/5] RUN pip install -r req.txt
⇒ [5/5] COPY . .
⇒ exporting to image
⇒ ⇒ exporting layers
⇒ ⇒ writing image sha256:9d346f9501aae69f3a8b5e9a76ffa02d5d65f28e004f7acb4553ceedbc3f051b
⇒ ⇒ naming to docker.io/library/falsk:2.0

└─(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab3]
└─$ la
app.py  Dockerfile  req.txt

└─(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab3]
└─$ cat Dockerfile
FROM python:3.10-slim

WORKDIR /app

COPY req.txt /app

RUN pip install -r req.txt

COPY . .

EXPOSE 8080

CMD ["python","app.py"]

└─(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab3]
└─$ sudo docker build -t flask-image .
[+] Building 1.1s (10/10) FINISHED
⇒ [internal] load build definition from Dockerfile
⇒ ⇒ transferring dockerfile: 169B
⇒ [internal] load metadata for docker.io/library/python:3.10-slim
⇒ [internal] load .dockerignore
⇒ ⇒ transferring context: 2B
⇒ [1/5] FROM docker.io/library/python:3.10-slim@sha256:2a8c7d9efdae9b636de80a187a4ce31374ffb3883a7be06232d25a96e8538bf4
⇒ [internal] load build context
⇒ ⇒ transferring context: 221B
⇒ CACHED [2/5] WORKDIR /app
⇒ CACHED [3/5] COPY req.txt /app
⇒ CACHED [4/5] RUN pip install -r req.txt
⇒ [5/5] COPY . .
⇒ exporting to image
⇒ ⇒ exporting layers
⇒ ⇒ writing image sha256:cdc77d44836d3cc2b578cc6d2f37d75dbc8c869f2998153d35ea12d5ae355a0
⇒ ⇒ naming to docker.io/library/flask-image

└─(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab3]
└─$ sudo docker run -it -p 8080:8080 --name flask-cont flask-image
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.18.0.2:5000
Press CTRL+C to quit
^C

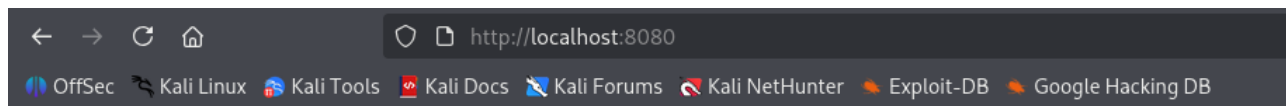
└─(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab3]
└─$ docker ps
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Get "http://x2Fvarx2Frunx2Fdocker.sock/v1.51/containers/json": dial unix /var/run/docker.sock: connect: permission denied

└─(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab3]
└─$ sudo docker ps
CONTAINER ID   IMAGE      COMMAND                  STATUS    PORTS       NAMES
└─(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab3]
└─$ sudo docker run -it -p 8080:5000 --name flask-cont flask-image
docker: Error response from daemon: Conflict. The container name "/flask-cont" is already in use by container "264f1ba67203cf2f478ed3bb88d405aaba6a7f09dae2a70d7e576fcfe3d17c0". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information

└─(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab3]
└─$ sudo docker rm flask-cont
flask-cont

└─(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab3]
└─$ sudo docker run -it -p 8080:5000 --name flask-cont flask-image
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.18.0.2:5000
Press CTRL+C to quit
172.18.0.1 - - [05/Nov/2025 15:14:11] "GET / HTTP/1.1" 200 -
```



Hello, Flask is running inside Docker!